

Hackathon

Problem statement

Development of Integrated, Affordable Solutions for Wastewater Treatment plants to Overcome Heavy-Metal Inhibition, Enable Microbial Activity Sensing, Enhance RO Organic-Load Tolerance, and Achieve Scalable RO-Reject Management.

Jointly hosted by

Ratan Tata Innovation Hub (RTIH)

Background & Context

Municipal and industrial wastewater systems are undergoing rapid change driven by rising contaminant variability, energy-intensive operations, and increasing water-reuse targets. Heavy metals commonly appearing in municipal wastewater inhibit microbial activity and demand larger bioreactor volumes, raising both infrastructure and sludge-management burdens.

At the same time, current aeration systems rely primarily on DO (Dissolved Oxygen) sensors, which do not reflect actual microbial biological activity—resulting in energy losses and sub-optimal sludge characteristics. Emerging biological markers and sensing platforms offer major improvement potential.

Wastewater treatment plants that depend on membrane separation technologies face limits due to organic loading, while RO (Reverse Osmosis) reject streams remain costly and difficult to manage across sectors.

Silica breakthrough in ultrapure water systems for semiconductor fabs leads to sudden spikes in silica concentration at polishing stages, impacting wafer yield and tool performance.

Challenge Overview

The Hackathon invites startups, innovators, institutions, and technology providers to design and develop solutions across wastewater operations and high-purity water systems.

1. **Bioreactor inhibition due to heavy metals** — The occurrence of heavy metals in municipal wastewater has become increasingly common across several cities. These

contaminants inhibit microbial activity in bioreactors, necessitating higher bioreactor volumes to maintain the required conversion efficiencies. Additionally, heavy metals accumulate in sludge, creating significant challenges for sludge treatability and safe disposal.

2. **Real-time microbial activity monitoring** — Real-time monitoring of microbial activity in wastewater reactors is essential for optimizing sludge and aeration. Currently, DO (Dissolved Oxygen) remains the only widely used sensor— leaving a major gap in tracking true biological activity.
3. **Functionalized RO membranes** — High organic loads restrict RO use in industrial wastewater, requiring COD <50 ppm or TOC <20 ppm. Advancing membrane surface functionalization and understanding organic–membrane interactions can broaden RO applicability. Building structure–activity insights will be key to designing membranes that perform under higher organic loads.
4. **RO (Reverse Osmosis) reject mitigation/valorization technologies** — RO is increasingly vital for water reuse and high-quality treatment, but RO reject management remains a critical bottleneck. Thus, cost-effective and scalable reject-mitigation or valorization technologies need to be developed to enhance the process.
5. **Silica breakthrough in ultrapure water systems for semiconductor fabs** - leads to sudden spikes in silica concentration at polishing stages, impacting wafer yield and tool performance. Existing RO, EDI (Electrodeionization), and resin systems fail to adequately reject monomeric (reactive) silica, creating a need for proactive, speciation-agnostic treatment solutions.

I. Challenge Vision and Rationale

Overarching Goal

To develop advanced, cost-effective wastewater and membrane-treatment technologies that improve biological stability, enhance membrane applicability, and enable sustainable RO-reject management. The overarching aim is to expand treatment capability while reducing plant footprint, energy demand, and dependence on narrow operating constraints across diverse water-treatment scenarios.

Rationale

Wastewater plants face persistent challenges: heavy-metal inhibition of bioreactors, lack of real-time microbial activity indicators, strict organic-load limits for RO operation, and unresolved RO-reject disposal issues. Addressing these bottlenecks is essential for improving conversion efficiency, reducing OPEX, enabling wider RO deployment, and ensuring scalable water-reuse solutions.

Core Objectives

1. **Remove heavy-metal inhibition** while preserving sludge quality and meeting safe concentration limits.
2. **Enable real-time microbial activity sensing** to optimize aeration and sludge performance.
3. **Develop functionalized RO membranes** for stable organic removal under higher loads.
4. **Create scalable RO-reject reduction or valorization solutions** to improve overall RO feasibility.
5. **Develop proactive, speciation-agnostic treatment solutions to prevent silica breakthroughs** in semiconductor ultrapure water systems

II. Targeted Participants and Eligibility

The challenge is open to participants, including:

Category	Description
Startups (Early to Growth Stage)	Industrial electronics, robotics, and power startups with scalable solutions
Independent Innovators	Individual developers with functional prototypes or algorithms
Academic & Research Institutions	Universities, labs, and incubators working on electronics and automation technologies
Technology Companies	Firms with existing platforms or components ready for customization and deployment

III. Scope of Solution

Participants are expected to design solutions covering any of the following functional scope:

Core Functional Requirements (Indicative areas: Non-Exhaustive)

1. **Bioreactor inhibition due to heavy metals**— Evaluate any physical, physico-chemical, electro-chemical, or biological approach that helps in removal of heavy metal impact on microbial inhibition in bioreactors (both aerobic and anaerobic) at a reduced cost.
2. **Real-time microbial activity monitoring** — Develop the ability to detect real-time microbial activity levels. Influence and optimize sludge characteristics by using microbial activity as a control parameter.

3. **Functionalized RO membranes** — Develop lab-scale membranes that enable separation of organic molecule with min. pressure loss over prolonged operating period.
4. **RO reject mitigation/valorization technologies** — Explore solutions of any type (solar-driven, electro-chemical, oxidative, precipitative, etc.) that can reduce, treat, or valorize RO reject. Validate the technology through laboratory testing and pilot-scale demonstrations
5. **Silica breakthrough in semiconductor industry** — Design and validate technologies that enable interlocking mechanisms for silica control, demonstrate capability to handle different silica forms, and pilot solutions for semiconductor fabs.

Participants are free to suggest focus areas beyond the above but aligned to the core objectives.

IV. Evaluation and Selection Criteria

Proposals will be evaluated by an expert committee based on the following criteria:

Evaluation Parameter	Weightage
Innovation & Originality	20
Technical Completeness & Scalability	10
User Experience	20
Real-World Applicability & Cost Effectiveness	40
Documentation & Demo Quality	10
Total	100

V. Rewards & Adoption Pathway

VI. Expected Deliverables

Participants must submit:

- Working prototype or simulation
- Source code and setup documentation
- Use-case demonstration aligned to problem statement

- System architecture diagram
- Demo video or live presentation.

VII. Goal & Expected Impact

Aims to:

- Enable more resilient and compact wastewater systems
- Improve process stability and reduce aeration costs
- Expand RO applicability across challenging wastewaters
- Enhance sustainability and economics of RO-based treatment
- Proactively control silica in semiconductor ultrapure water systems